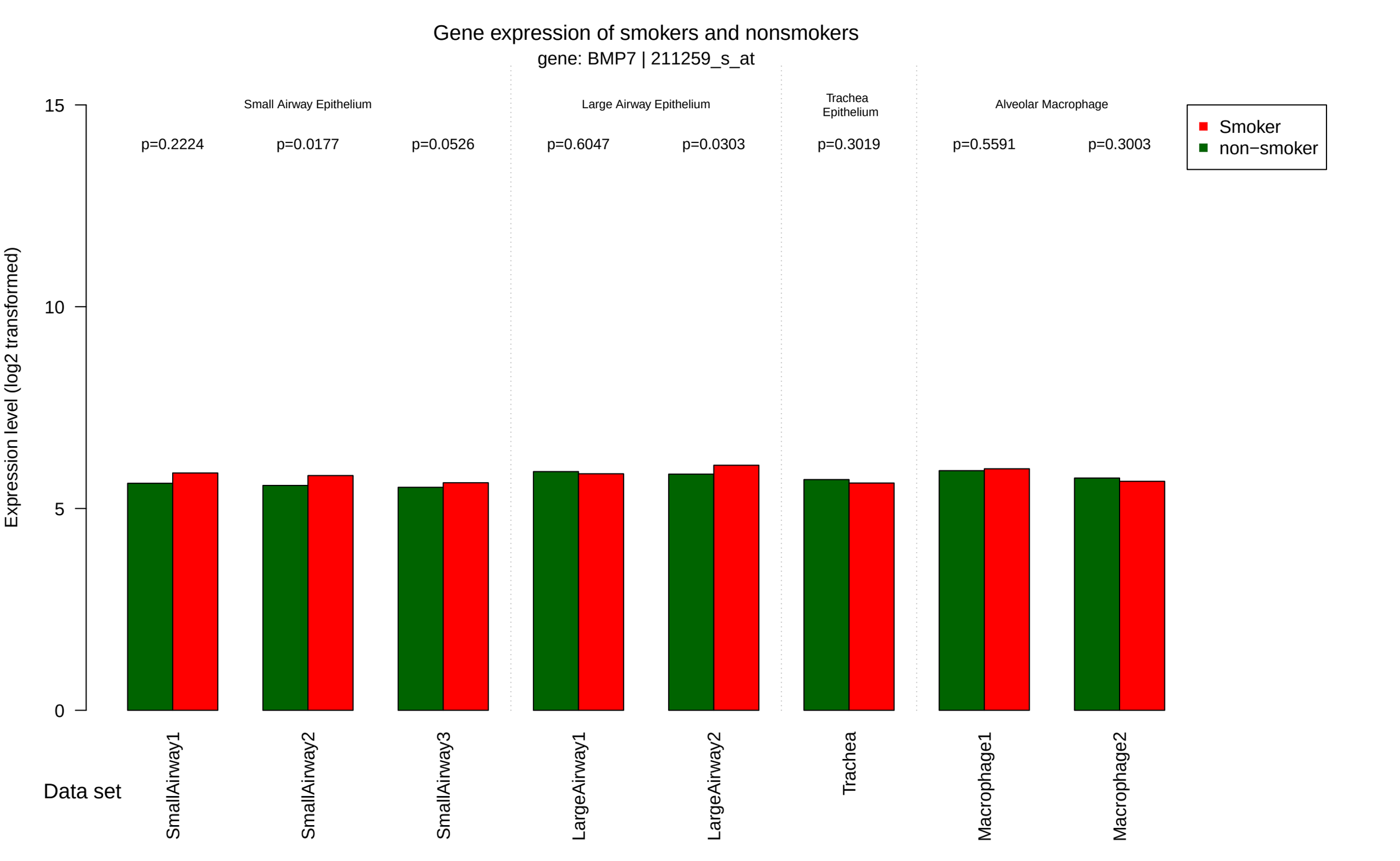
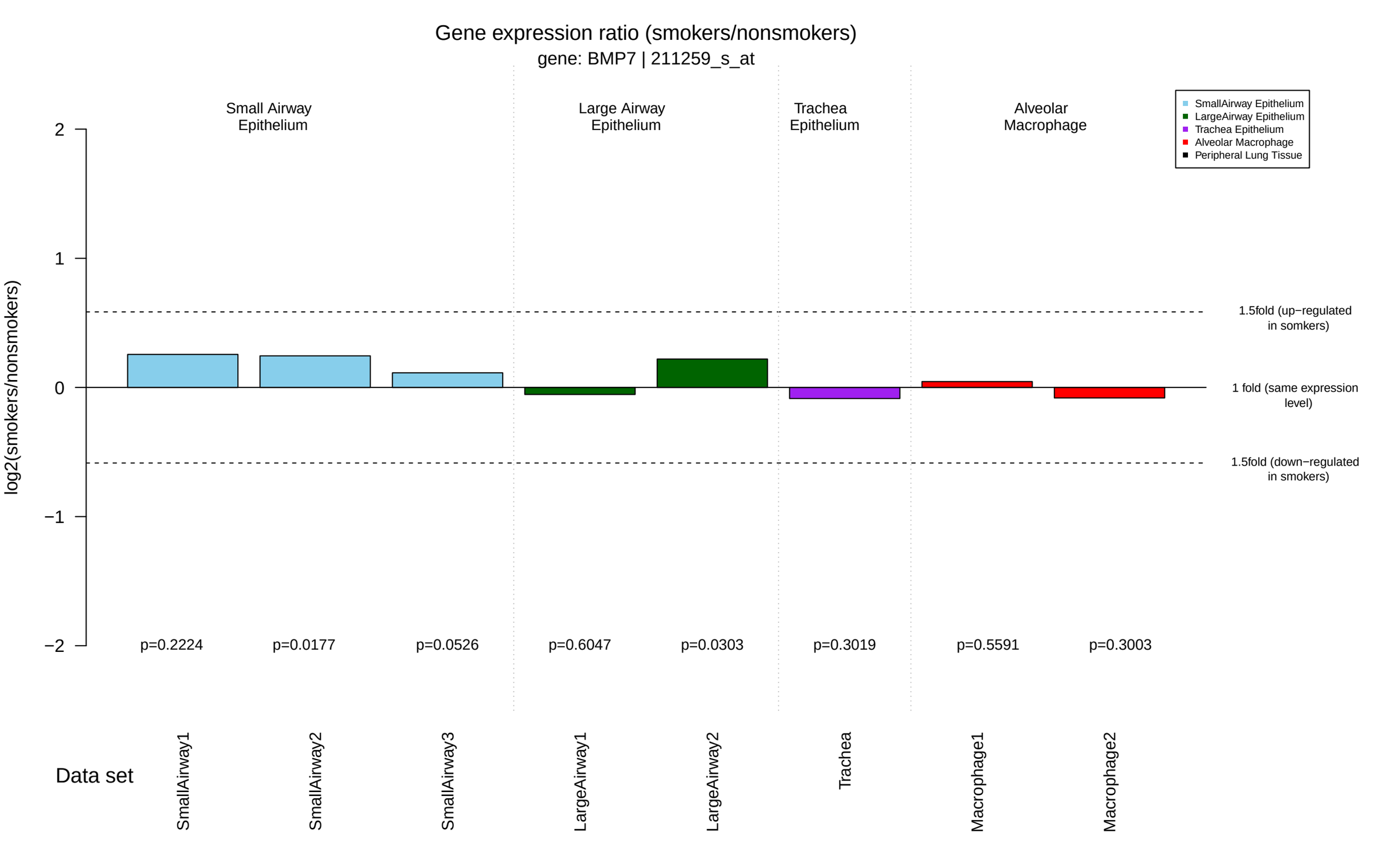
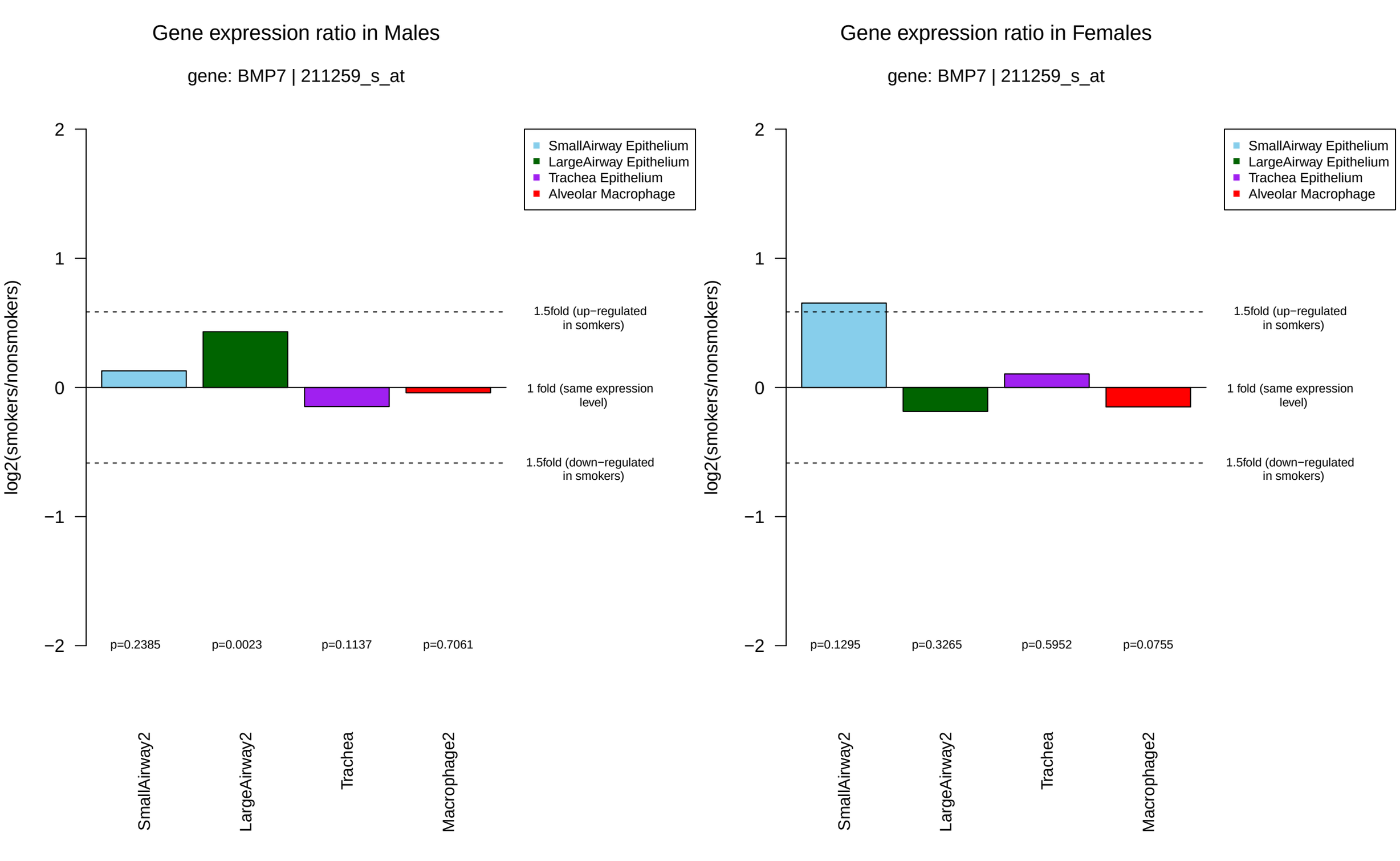
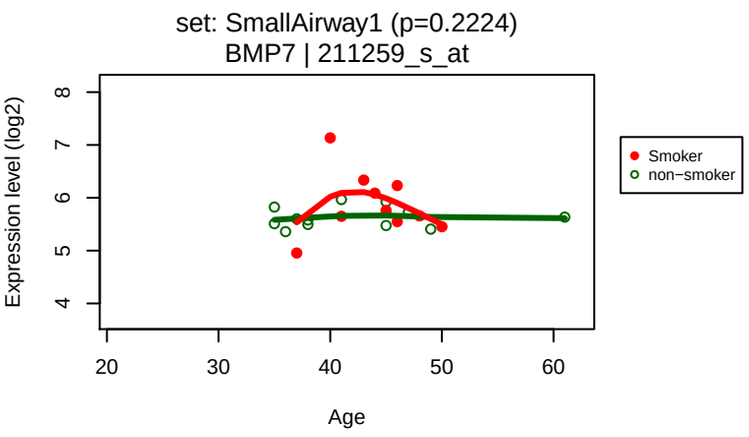
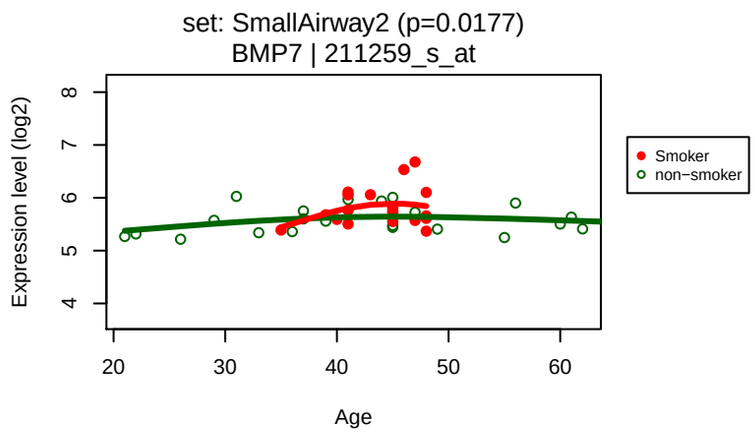
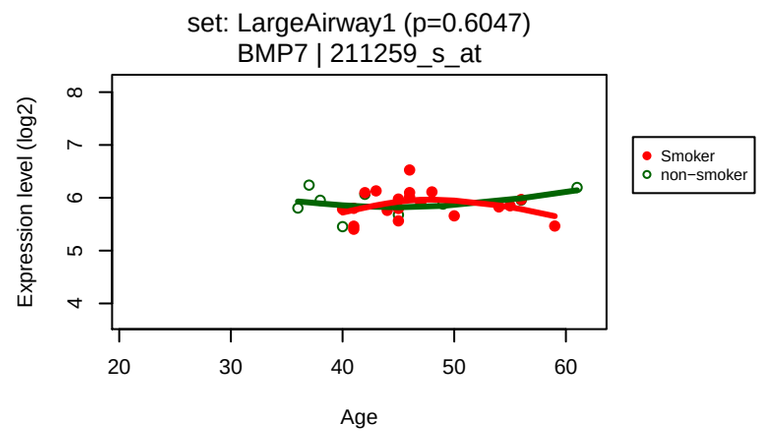

Gene expression of smokers and nonsmokers
gene: BMP7 | 211259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2224
p=0.0177
p=0.0526
p=0.6047
p=0.0303
p=0.3019
p=0.5591
p=0.3003
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

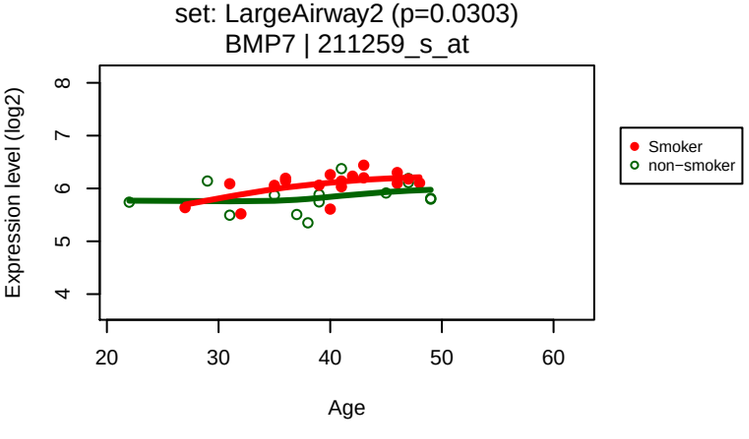

Gene expression ratio (smokers/nonsmokers)
gene: BMP7 | 211259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2224
p=0.0177
p=0.0526
p=0.6047
p=0.0303
p=0.3019
p=0.5591
p=0.3003
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

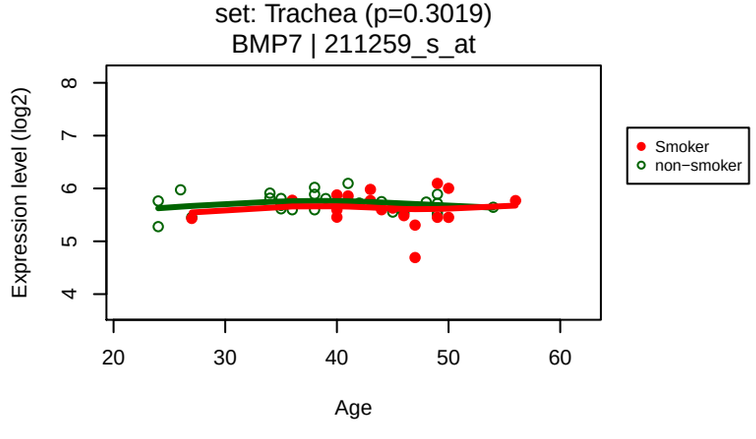

Gene expression ratio in Males
gene: BMP7 | 211259_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2385
p=0.0023
p=0.1137
p=0.7061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP7 | 211259_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1295
p=0.3265
p=0.5952
p=0.0755
SmallAirway2
LargeAirway2
Trachea
Macrophage2

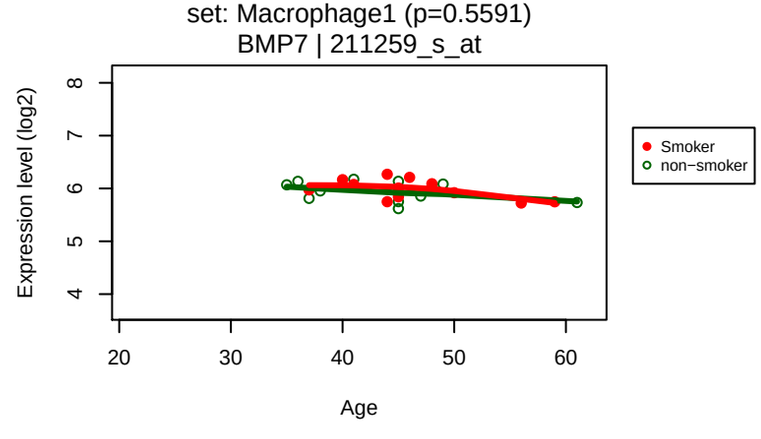

set: SmallAirway1 (p=0.2224)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker

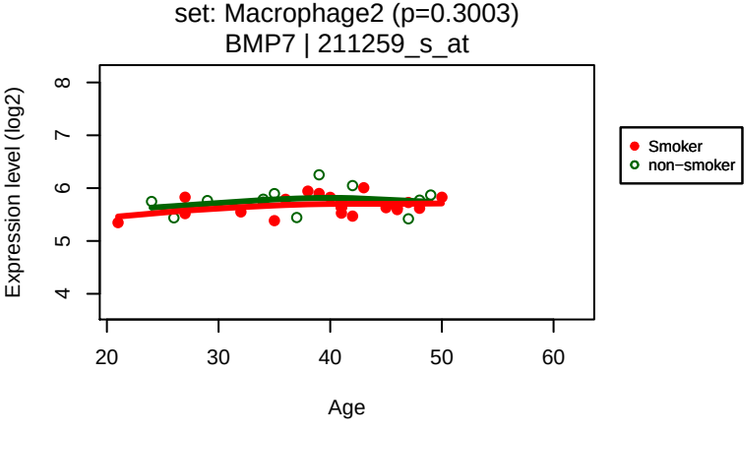

set: SmallAirway2 (p=0.0177)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker

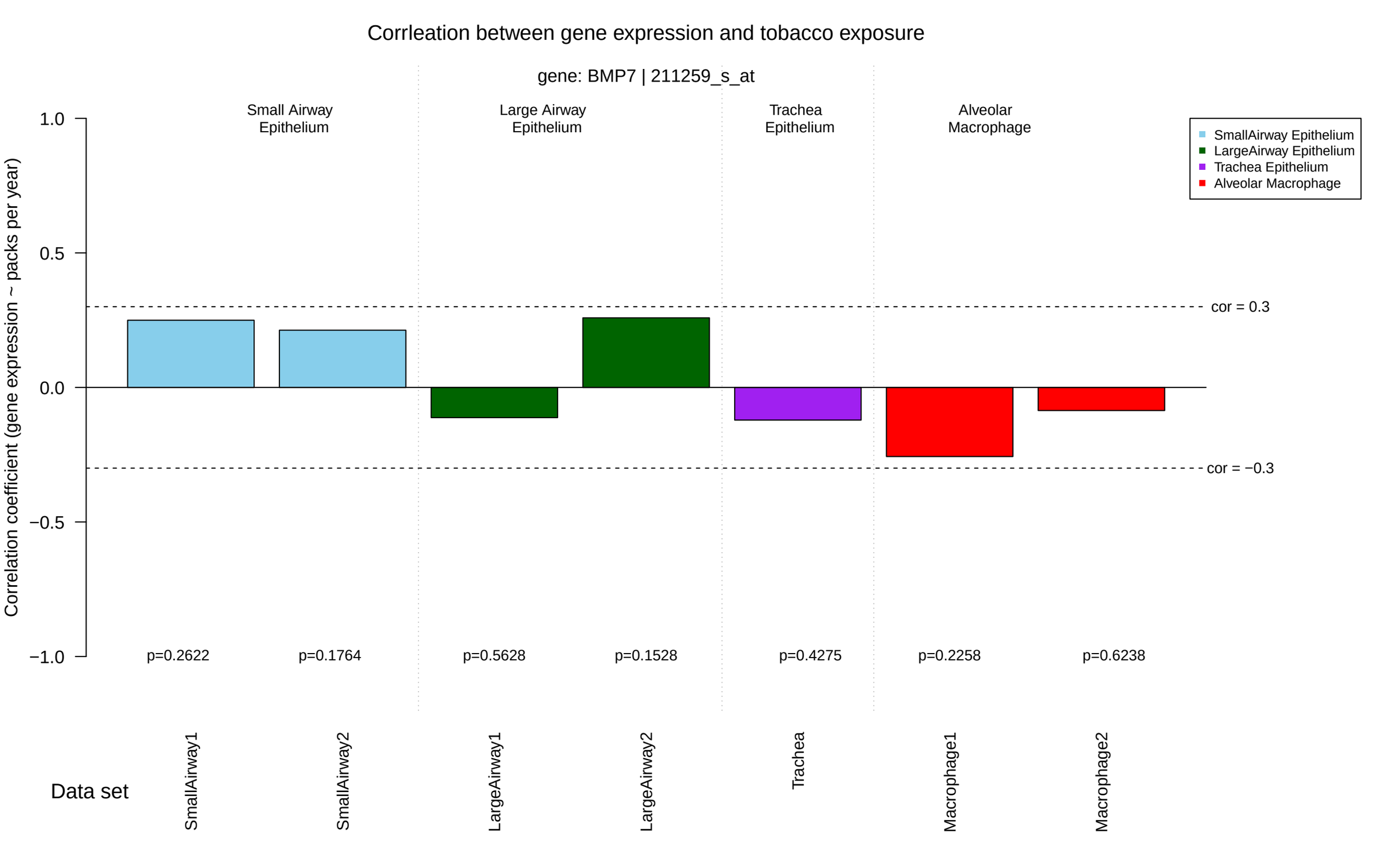

set: LargeAirway1 (p=0.6047)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0303)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3019)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5591)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3003)
BMP7 | 211259_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP7 | 211259_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2622
p=0.1764
p=0.5628
p=0.1528
p=0.4275
p=0.2258
p=0.6238
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2